Key

Style 5

- Truth and Nil 6
- Arithmetic 16
- Constants and Chance 38
- Number Theory 48
- Polynomials and Roots 55
- Matrices 62
- Arrays 70
- Boxes 85
- Sets and Membership 96
- Trains 105
- Functions 123
- Errors 132
- Classes and Properties 143
- The Virtual 153

Dungeon 159

- Places 160
- Classes 164
- Objects 170
- Entities 175
- Food 178
- Species inheritance 197

Index 239

- English Index 240
- Glyph Index 247

Contents

Style 5

- Truth and Nil 6
  - Niladic 6
  - Monadic 8
  - Dyadic 10
- Arithmetic 16
  - Niladic 16
  - Monadic 20
  - Dyadic 31
- Constants and Chance 38
  - Niladic 38
  - Monadic 42
  - Dyadic 45
- Number Theory 48
  - Monadic 48
  - Dyadic 51
- Polynomials and Roots 55
  - Monadic 55
  - Dyadic 58
- Matrices 62

Niladic 62
Monadic 64
Dyadic 67
Arrays 70
Niladic 70
Monadic 73
Dyadic 80
Boxes 85
Niladic 85
Monadic 87
Dyadic 91
Sets and Membership 96
Niladic 96
Monadic 98
Dyadic 101
Trains 105
Niladic 105
Monadic 108
Dyadic 120
Functions 123
Monadic 123
Dyadic 128
Errors 132
Niladic 132
Monadic 138
Classes and Properties 143
Niladic 143
Monadic 146

Dyadic 149
The Virtual 153
Niladic 153
Monadic 156
Dungeon 159
Places 160
Classes 164
Objects 170
Entities 175
Food 178
Species inheritance 197
Index 239
English Index 240
Glyph Index 247

Style

Truth and Nil: Niladic

Wuxia

俠

The Hero

U+4FA0

Nil



The empty set — quiet failure value; absorbs; Nil⊡function → Nil

U+2205

Truth

⊤

Verum — first-class truth; no true anymore

U+22A4

Truth and Nil: Monadic

Invert

~

Invert — monadic only (tilde). Scalar: ~ 0 → 1, ~ 1 → 0 (user wording to confirm). Unary scalar verbs operate on array back ranks (new 2026-10-11)

U+007E

Not

¬

Not — monadic boolean not: ¬ Truth yields Nil, ¬ Nil yields Truth (new 2026-10-14)

U+00AC

Congruent

≡

Ambivalent: monadic binds train to lhs⊡function

U+2261

Truth and Nil: Dyadic

Less Than



Less Than — dyadic < A B yields Truth when A < B, Nil otherwise (new 2026-10-14)

U+003C

Greater Than



Greater Than — dyadic > A B yields Truth when A > B, Nil otherwise (new 2026-10-14)

U+003E

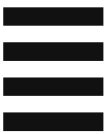
Congruent



Ambivalent: dyadic ≡ A B yields Truth when A and B have the same binding (same location in the system tables), Nil otherwise

U+2261

IsAt



IsAt — dyadic ≣ X Y yields Truth when X and Y share the same pointer address, Nil otherwise (new 2026-10-14)

U+2263

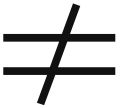
Equality



Equality — dyadic = A B yields Truth when A = B, Nil otherwise (new 2026-10-14)

U+003D

Unequal



Unequal — dyadic ≠ A B: sugar for ¬ = A B (boolean Not of Equality) (new 2026-10-14)

U+2260

Less Or Equal



Less Or Equal — dyadic ≤ A B yields Truth when A ≤ B, Nil otherwise (new 2026-10-14)

U+2264

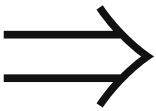
Greater Or Equal



Greater Or Equal — dyadic ≥ A B yields Truth when A ≥ B, Nil otherwise (new 2026-10-14)

U+2265

Implies



Implies — logical implication; ∅ ⇒ ∅ → ⊤

U+21D2

forall



Train quantifier — every element must evaluate non-nil. Idiom: dyadic NAND is ¬ ∀ 𝕀𝕀 X Y

U+2200

Approx



Approx — dyadic ≈ A B yields Truth when|A - B|is within 1e-6, Nil otherwise (new 2026-10-14)

U+2248

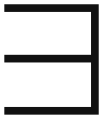
UnApprox



UnApprox — dyadic ≉ A B: sugar for ¬ ≈ A B (boolean Not of Approx) (new 2026-10-14)

U+2249

Forsome



Existential — at least one element evaluates non-nil

U+2203

Fornone



Not-forsome — defined as ~Forsome

U+2204

Arithmetic: Niladic

Signum / Plus



Ambivalent: Niladic + yields 0.
(Was ∓ U+2213, retired 2026-10-12

U+002B

Negate / Subtract

Niladic - yields 0 (new 2026-10-11

U+002D

Plus (Complex)



Plus (Complex), aka Sum Complex, Add Complex —; Niladic: sugar for ℿ 0 0, the complex zero (new 2026-10-14)

U+2A22

Times (Complex)



Times (Complex) —; Niladic: sugar for ℿ 1 0, the complex one (new 2026-10-14)

U+2A30

Minus (Complex)



Minus (Complex) —; Niladic: sugar for ⨿ 0 0, the complex zero (new 2026-10-14)

U+235B

Times (Polar)



Times (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)

U+2A02

Divide (Complex)



Divide (Complex) —; Niladic: sugar for ⨿ 1 0, the complex one (new 2026-10-14)

U+2A2C

Divide (Polar)



Divide (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)

U+2A38

Factorial

!

Factorial —; Niladic ! yields 1 (new 2026-10-12)

U+0021

Percent

%

Percent —; Niladic % is 0 (new 2026-10-12

U+0025

Square



Square — ⊠ X is X squared.; Niladic ⊠ yields 1, derived from Multiply (new 2026-10-12)

U+22A0

Multiply



*Multiply —; Niladic × yields empty array. Replaces Star * U+002A (new 2026-10-12; boxed forms 2026-10-12*

U+2A2F

Arithmetic: Monadic

Signum

+

Ambivalent: monadic Signum (less than zero yields −1, greater than zero yields 1, zero yields 0; back-rank scalar unary)

U+002B

Negate

-

Negate (monadic - X, unary minus)

U+002D

Conjugate

⊶

Conjugate — monadic ⊶ X: complex conjugate; negates the imaginary part, preserving orientation (on [r,i] yields [r,−i]; on [r;i] yields [r;−i]) (new 2026-10-15)

U+22B6

Double

+2

Double — monadic +⁼ X yields 2 × X (new 2026-10-15)

U+2A27

Sum

∑

Sum — sugar for reduce-addition: ∑ X means / + X (redefined 2026-10-12)

U+2211

Square Root

√

Square Root — monadic √ X yields the square root of X (new 2026-10-15)

U+221A

Plus (Complex)

⨁

Plus (Complex), aka Sum Complex, Add Complex —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.

U+2A22

Times (Complex)

⨂

Times (Complex) —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.

U+2A30

Minus (Complex)



Minus (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.

U+235B

Times (Polar)



Times (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.

U+2A02

Divide (Complex)



Divide (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.

U+2A2C

Divide (Polar)



Divide (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.

U+2A38

Percent

%

Percent —; monadic % X is absolute value, a back-rank operator.; monadic redefined 2026-10-12)

U+0025

Star

*

*Star — exponentiate: monadic * X is 1, because the missing exponent Y is assumed 0, giving X^0 (new 2026-10-12). Idiom: X-th root of Y is * Y ÷ 1 X*

U+002A

Quotient

÷

Quotient —; Monadic ÷ X: if X is a square array, yields the matrix inverse; else reciprocal, a back-rank scalar operator (new 2026-10-12)

U+00F7

Multiply

×

Multiply —; Monadic × X yields X when X numeric; on boxed X, catenates once.; monadic redefined 2026-10-12)

U+2A2F

Ceiling



Ceiling —; monadic ⌈ X rounds up
(from APL; new 2026-10-12)

U+2308

Floor



Floor —; monadic ⌊ X rounds down
(from APL; new 2026-10-12)

U+230A

Sine



Sine (new 2026-10-12)

U+25DD

Cosine



Cosine (new 2026-10-12)

U+25DE

Arccosine



Arccosine (new 2026-10-12)

U+25F6

Arcsine



Arcsine (new 2026-10-12)

U+25F7

Tangent



Tangent (new 2026-10-12)

U+2AD7

Arctangent



Arctangent (new 2026-10-12)

U+2AD8

Hyperbolic Sine



Hyperbolic Sine (new 2026-10-12)

U+2AA6

Hyperbolic Cosine



Hyperbolic Cosine (new 2026-10-12)

U+2AA7

Hyperbolic Arcsine



Hyperbolic Arcsine (new 2026-10-12)

U+2ABD

Hyperbolic Tangent



Hyperbolic Tangent (new 2026-10-12)

U+2AD1

Natural Log



Natural Log —; monadic ⍁ X is ln X.; Monadic ⍁ on a polar 2×1 yields the rectangular complex pair [ln r, θ].

U+237C

Antilog



Antilog — sugar for Star: ⍋̇ X is e raised to the X. Monadic ⍋̇ on a rectangular complex pair yields the polar 2×1 [e^a; b] (new 2026-10-12; glyph was 𝒆 U+1D58A until 2026-10-14)

U+29CA

Hyperbolic Arctangent



Hyperbolic Arctangent (new 2026-10-12)

U+2A4D

Hyperbolic Arccosine



Hyperbolic Arccosine (new 2026-10-12)

U+2ABE

Factorial

!

Factorial —; monadic ! X is X factorial

U+0021

Fourier



Fourier — monadic F X: FFT of X (new 2026-10-15)

U+03DC

InverseFourier



InverseFourier — monadic ⅁ X: inverse FFT of X (new 2026-10-15)

U+2132

Base Ten Log



Base Ten Log — ⍙ X is log base 10 of X, sugar for ⍦ 10 X (new 2026-10-12)

U+2A5F

Magnitude



Magnitude — monadic ⍫X: absolute value of X (new 2026-10-16)

U+2357

Arithmetic: Dyadic

Plus



Ambivalent: dyadic + X Y yields X + Y (addition).; dyadic 2026-10-15)

U+002B

Subtract

Subtract (dyadic - X Y yields X - Y, first minus second), the subtract that works with the / right-fold Reduce.; dyadic 2026-10-12)

U+002D

Plus (Complex)



Plus (Complex), aka Sum Complex, Add Complex —; dyadic: sum of complex numbers (1×2 rectangular), per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before adding.

U+2A22

Times (Complex)



Times (Complex) —; dyadic: product of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before multiplying.

U+2A30

Minus (Complex)



Minus (Complex) —; dyadic: difference of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before subtracting.

U+235B

Times (Polar)



Times (Polar) —; dyadic: product of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes multiply, angles add; if either or both X and Y are scalar, promote to polar 2×1 before multiplying.

U+2A02

Divide (Complex)



Divide (Complex) —; dyadic: quotient of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before dividing.

U+2A2C

Divide (Polar)



Divide (Polar) —; dyadic: quotient of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes divide, angles subtract; if either or both X and Y are scalar, promote to polar 2×1 before dividing.

U+2A38

Percent

%

Percent —; dyadic % X Y yields the remainder of X divided by Y

U+0025

Star

*

*Star — exponentiate: dyadic * X Y is X to the power Y (left=base, right=exponent)*

U+002A

Quotient

÷

Quotient —; dyadic ÷ X Y divides X by Y, yielding a real; division by zero yields infinity, except 0/0 which yields Nil.

U+00F7

Multiply

×

Multiply —; dyadic multiplies its arguments; when X is boxed, × X Y catenates X repeated Y times.

U+2A2F

Factorial

!

Factorial —; dyadic ! X Y is the binomial coefficient Y choose X.

U+0021

Ceiling

⌈

Ceiling —; dyadic ⌈ X Y is maximum

U+2308

Floor

⌊

Floor —; dyadic ⌊ X Y is minimum

U+230A

Natural Log



Natural Log —; Dyadic ⍦ X Y is log base X of Y, sugar for ÷ ⍦ Y ⍦ X (new 2026-10-12)

U+237C

Square



Square — ⊠ X is X squared.; Dyadic ⊠ X Y squares X Y times.

U+22A0

Constants and Chance: Niladic

Half

½

Half —; niladic: 0

U+00BD

Pi

π

Pi — niladic π: value of pi (new 2026-10-14)

U+03C0

Bell

𝔅

Bell —; niladic: random point in (-1, 1), exclusive

U+1D56D

Real

ℜ

Real — niladic: random real number between 0 and 1; sugar for ½ + 1 𝔅 (new 2026-10-13)

U+1D57D

Gamma



Gamma — niladic γ: Euler-Mascheroni constant (new 2026-10-14)

U+03B3

Golden



Golden — niladic φ: golden mean (new 2026-10-14)

U+03C6

Euler



Euler — niladic ℇ: Euler's constant (new 2026-10-14)

U+2107

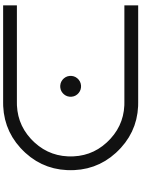
Dice



Dice —; niladic: single 6-sided die (1-6)

U+1D56F

Pick



Pick —; niladic: Nil

U+2A03

Constants and Chance: Monadic

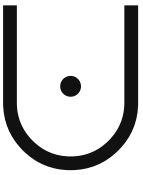
Half

½

Half —; monadic ½ X: X divided by 2; array → sugar for ⊞ (⍴ X); train → floor(length/2) index; dictionary → Nil (new 2026-10-13)

U+00BD

Pick



Pick —; monadic ⌣⊙X: scalar → same scalar; train → use Dice of size train to select element; array → use Dice of sides from shape to select element; dictionary → random {key:value} pair (new 2026-10-13)

U+2A03

Bell

𝔅

Bell —; monadic 𝔅 μ: bell curve point with mean μ (μ in (-1,1))

U+1D56D

Dice

𝔇

Dice —; monadic 𝔇 X: single X-sided die (1-X)

U+1D56F

Deal



Deal — given a scalar, works as expected; given an array, fills the entire back rank with limited values. Concrete examples and summary TBD (new 2026-10-11)

U+003F

Constants and Chance: Dyadic

Bell



Bell —; dyadic 𝔅 μ σ: bell curve point with mean μ and stddev σ (σ ≤ 1/3) (new 2026-10-13)

U+1D56D

Dice



Dice —; dyadic 𝔇 X N: N X-sided dice; all sugar via 𝔑 (hence 𝔅) (new 2026-10-13)

U+1D56F

Number Theory: Monadic

Divisible



Divisible —; monadic ∣ X yields Truth when X is divisible by 2, Nil otherwise

U+23A2

Indivisible



Indivisible —; monadic ∤ X: sugar for ¬ ∣ X (the Odd test)

U+2AEE

Primes



Primes — monadic ℙ X yields the prime factors of X, per J's q:. (new 2026-10-18)

U+1D4AB

Prime



Prime — monadic 𝔓 N yields the Nth prime, per J's p: (zero-indexed: 𝔓 0 is 2). (new 2026-10-18)

U+1D513

Prime



Prime — monadic ⌽ X: sugar for ⍭⍘ X ⍳ 2 ⌊ √ X ⋮ ≤ X 1 (new 2026-10-15)

U+23C0

Factor



Factor — factorization verb (new 2026-10-16)

U+27D2

Combinatoric



Combinatoric —; monadic ## X: factorial of X

U+2A69

Number Theory: Dyadic

LCM



LCM — dyadic ⨶ X Y: least common multiple of X and Y (new 2026-10-16)

U+238A

Divisible



Divisible —; dyadic ∣ X Y yields Truth when X is divisible by Y, Nil otherwise (new 2026-10-15)

U+23A2

GCD



GCD — dyadic ⫻ X Y: greatest common divisor of X and Y (new 2026-10-16)

U+2980

Indivisible



Indivisible —; dyadic ⫲ X Y: sugar for ¬ ∣ X Y (boolean Not of Divisible) (new 2026-10-15)

U+2AEE

Combinatoric



Combinatoric —; dyadic ⋕ X Y: binomial X choose Y (new 2026-10-16)

U+2A69

Polynomials and Roots: Monadic

NextLocal



NextLocal — the Roots division step: given P, yields [R prime, P prime] where R prime is Local(P) and P prime is P deflated by synthetic division of R prime. (new 2026-10-19)

U+2113

Roots



Roots — sugar: ℞ P is ℜ ℓ P (Reveal NextLocal over P); yields all polynomial roots. (new 2026-10-18; factored via Reveal 2026-10-19)

U+211E

Differentiate



Differentiate — train verb ∂ T yields symbolically differentiated version of train T of coefficients, multiplying constants as needed; same order as Polynomial (new 2026-10-13)

U+2202

Integrate



Integrate — train verb ∫ T yields symbolically integrated version of train T of coefficients, dividing constants as needed; same order as Polynomial (new 2026-10-13)

U+222B

Solve



Solve — monadic & A: array → vector of polynomial coefficients; missing input row is all zeroes (new 2026-10-13)

U+0026

Optimize



*Optimize —; monadic 𝔅 V: vector → polynomial; searches free variable with delta = 0.01 * domain, epsilon defaults to 1e-6*

U+214B

Local



Local — monadic 𝔴 V is 𝔅 𝕀𝕀𝕀 ℒ & V (FixedPoint Optimize Triple Solve V); approximates best x for a local minimal of V (new 2026-10-13)

U+1D50F

Polynomials and Roots: Dyadic

Reveal



Reveal — intrinsic. Dyadic ℜ P Y yields the successive application of P upon Y as Y shrinks to Nil. Epsilon from P has two elements: an alpha to apply P to next Y, and the next Y. (new 2026-10-18)

U+211C

Optimize



Optimize —; dyadic ϐ V ε: recurse into smaller domains until change in c < ε (new 2026-10-13)

U+214B

Integer Division



Integer Division — dyadic ⫽ X Y divides X by Y, yielding an integer (replaces Slash / integer division from manual p.19; / is now Reduce). Vector form: ⫽ V X where V is vector yields synthetic quotient of V by X (new 2026-10-14)

U+2AFD

Evaluate



Evaluate — dyadic 𝒇 X T evaluates train T of coefficients upon X (new 2026-10-13)

U+1D58B

FixedPoint



Fixed point — dyadic cø V guess: refine guess until epsilon < 1e-6 (max 100 steps); failure to converge returns infinity (new 2026-10-13)

U+AB44

Matrices: Niladic

Outer Product



Outer Product —; Niladic ⊗ yields 1 (new 2026-10-12)

U+2297

Inner Product



Inner Product —; Niladic · yields 1 (new 2026-10-12; matrix case 2026-10-14)

U+22C5

Multiply Matrices



Multiply Matrices —; Niladic ⋈ yields the 1×1 identity matrix (new 2026-10-14)

U+22C8

Matrices: Monadic

Outer Product



Outer Product —; Monadic ⊗ X yields X.

U+2297

Inner Product



Inner Product —; Monadic · X yields X.

U+22C5

Multiply Matrices



Multiply Matrices —; Monadic ⋈ X yields X.

U+22C8

L Factor



L Factor — monadic: the L (lower triangular, 1s on diagonal) factor of the LU decomposition of a rank-2 array (new 2026-10-14)

U+1D577

Determinant



Determinant — monadic ⌹X: determinant of square matrix X (new 2026-10-16)

U+233C

U Factor



U Factor — monadic: the U (upper triangular) factor of the LU decomposition of a rank-2 array (new 2026-10-14)

U+1D580

Matrices: Dyadic

Outer Product



Outer Product —; dyadic ⊗ X Y yields the outer product matrix where entry [i,j] is × X[i] Y[j]; shape is the concatenation of the shapes of X and Y.

U+2297

Inner Product



Inner Product —; dyadic · X Y: for vectors X, Y, the dot product (sum of pairwise products); for matrices, the Frobenius inner product: a scalar, the sum of all pairwise products.

U+22C5

Multiply Matrices



Multiply Matrices —; dyadic ⋈ X Y: matrix multiplication, (⋈ X Y)[i;j] = · X[i;] (⍤ Y)[j;]; inner dimensions must agree, non-square fine.

U+22C8

Arrays: Niladic

Pee



Pee — Where⊡Pee yields Where [0, 1
(first two coordinates)

U+2119

Zee



Zee — the last position of a slice
(Where⊡Zee yields Where [-1)

U+2124

Exx



Exx — the zero position of a slice
(Where [0)

U+1D54F

Wye



Wye — the midposition of a slice

U+1D550

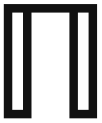
Extend



Extend —; niladic ⊢: sugar for ⊂ Nil

U+213E

Catenate



Catenate —; niladic ⊣: sugar for ⊂ Nil

U+213F

Grade Up



Grade Up — ⍋X yields the indices that sort X ascending. Niladic ⍋ yields 0 (from APL; new 2026-10-12)

U+234B

Grade Down



Grade Down — ⍒X yields the indices that sort X descending. Niladic ⍒ yields Nil (from APL; new 2026-10-12)

U+2352

Arrays: Monadic

Rank



Rank — monadic ⌗ X yields the number of dimensions in X; zero when scalar. (new 2026-10-18)

U+22D5

Permute



Permute —; monadic ⍨ X and dyadic X ⍨ Y work as in J's C.. Both modes intrinsic. (new 2026-10-19)

U+2A0A

Laminate



Laminate —; monadic ⍪ X: adds a leading singleton axis

U+2A5E

Anagram



Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)

U+1D49C

Enumerate

№

Enumerate — monadic № X: sugar for ⍕⍳≢⍴X; yields index-value pairs for X (new 2026-10-18)

U+2116

Reflect

÷

Reflect — monadic ÷X: sugar for ⍕⇋-⇋::X (new 2026-10-15)

U+223B

Count

76

⍳

U+2373

Kount

⍭

Kount — monadic ⍭B: takes a vector of T or Nil; returns a vector counting the T, turning each Nil to 0. Intrinsic. (new 2026-10-18)

U+2A23

⍳ Count

U+2373

Count — magical: when given a shape, fills an array matching shape. Scalar: behaves as normal. Array: fills array with next dimension limited by scalar from back rank — e.g. ⍳ 2 3 → rows of 0 1 2 (origin 0; last axis counts, outer axes replicate) (new 2026-10-11)

RavelRows



RavelRows — monadic ⍪ X: yields the rows of X as a vector (Antitable; new 2026-10-15)

U+22F2

Ravel



Ravel — monadic ⍎ X: flattens array X to vector, row-major (new 2026-10-14)

U+22F3

Reverse



Reverse —; monadic ⌽ X reverses elements along the last axis

U+233D

Transpose



Transpose — monadic ⍉ X reverses the order of axes (from APL; new 2026-10-12)

U+2349

Shape



Shape —; monadic ρ X: shape of X; scalar → 0; array → vector of dimensions; train → [1, length of train]; dictionary → [1, pairs present]

U+03C1

Extend



Extend —; monadic ⍮ X: Table X as in APL, adds trailing axis of length 1

U+213E

Catenate



Catenate —; monadic ⍪ X: catenates X to itself

U+213F

Rotate



Rotate —; monadic ⌽ X: rotate X toward origin by 1

U+29BD

Cardinality



Cardinality — "how big is this?"; scalar → -1; array → number of elements held; dictionary → number of pairs; train → length of train (new 2026-10-13)

U+2210

Center



Center — takes a shape, returns its integer halves expressed as a vector (e.g. [10, 6] → [5, 3]) (new 2026-10-13)

U+229E

Sight



Sight — takes 2D array, finds center, returns vector location of nearest non-zero scalar; ties pick one (new 2026-10-13)

U+2388

Arrays: Dyadic

Put



Put — ⫤⫥ X P A: value at coordinate P of array A becomes X (new 2026-10-18)

U+27DA

Permute



Permute —

U+2A0A

Laminate



Laminate —; dyadic ⩓ A B: yields [A; B]; when A and B are both scalars, yields A B (new 2026-10-17)

U+2A5E

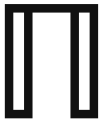
Anagram



Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)

U+1D49C

Catenate



Catenate —; dyadic ⎍ X Y: catenate X and Y as in APL; variadic ⎍ A ... Z: catenate A through Z left to right (new 2026-10-14)

U+213F

Take



Take — ↑ X Y takes the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)

U+2191

Drop



Drop — ↓ X Y drops the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)

U+2193

Reverse



Reverse —; dyadic ⌽ X Y rotates Y by X positions (from APL; new 2026-10-12)

U+233D

Shape

ρ

Shape —; dyadic ρ S X: reshape X to shape S (per APL) (new 2026-10-13)

U+03C1

Extend



Extend —; dyadic ⍋ X N: extends by N axes (new 2026-10-14)

U+213E

Rotate



Rotate —; dyadic ⍜ X Y: rotate X by Y (Y positive toward origin, negative away) (new 2026-10-16)

U+29BD

Boxes: Niladic

Box



Box —; Niladic ⊂ yields Truth
(from APL Enclose; new 2026-10-12)

U+2282

Unbox



Unbox —; Niladic ⊃ yields Nil
(from APL Disclose; new 2026-10-12)

U+2283

Summarize



Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; niladic ⊚ is the 1,1 vector containing zero.

U+229A

Boxes: Monadic

Box



Box — monadic ⧆ A: wraps A in a box so it travels as a single element (new 2026-10-17)

U+2256

Unbox



Unbox — monadic ⧈ A: opens the box, yielding its contents (new 2026-10-17)

U+2257

Catalogue



Catalogue —; monadic ⊚X: cartesian product of the boxed lists in vector X (new 2026-10-17)

U+2A37

Raze



Raze — monadic ⫻A: sugar for Ravel Map Unbox A; opens every box and joins the contents into one vector (new 2026-10-17)

U+2AF5

Box



Box —; monadic ⊂ X encloses X into a scalar box.

U+2282

Unbox



Unbox —; monadic ⊃ X yields X.

U+2283

Summarize



Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar

U+229A

Zip

90



U+22B0



Zip

`U+22B0`

Zip — takes a train of vectors and zips them: ⍮ V1 V2 ... Vn yields an m×n matrix (m = vector length, n = train length); the resulting width is the length of the train; a scalar in the train expands into a column to fit the others; a container in the train contributes only its origin scalar, recursively (the summary effect); mismatched lengths Nil-pad (consistent with Diction); zips along the back rank for higher-rank arrays; ⍮ 1 2 yields the column [1;2] (new 2026-10-12; recycles the retired AntiSignum glyph)

Boxes: Dyadic

Summarize

93



U+229A

Fetch



Fetch — dyadic ⊸ V X: applies V as search vector into X's items; follows path V down through nested boxes (new 2026-10-17)

U+22B8

Catalogue



Catalogue —; dyadic ⊗ A B: yields [[A0,B0] ... [A1,B1]] ... [[AN,B0] ... [AN,BN]], the outer product matrix of pairs

U+2A37

Link



Link — dyadic ⫢ X Y: sugar for Catenate Map Box X Y; boxes X and Y into a two-element list (new 2026-10-17)

U+2AE9



Summarize

U+229A

Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; Dyadic D ⊚ A: D (Direction) is a rank-1 array of coordinates; summarization starts at A[D], then proceeds as monadic; a scalar Direction s means start at (s, 0); Zip summarizes each train element (new 2026-10-12)

Box



Box —; Dyadic ⊂ X Y encloses X by Y levels; Y=0 leaves X unchanged.

U+2282

Unbox

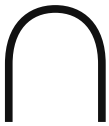


Unbox —; Dyadic ⊃ X Y picks element X from Y.

U+2283

Sets and Membership: Niladic

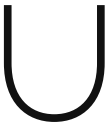
Intersection



Intersection —; niladic ∩: sugar for ⊂ Nil

U+2229

Union

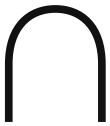


Union —; niladic ∪: sugar for ⊂ Nil

U+222A

Sets and Membership: Monadic

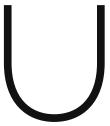
Intersection



Intersection —; monadic ∩ A: is A

U+2229

Union



Union —; monadic ∪ X: unique as in APL

U+222A

InOf



InOf — the curry of In: ⍨InOf ⍷ ∈ X; monadic predicate testing membership in X. (new 2026-10-18)

U+22F8

Where



Where — monadic ⍸Y: returns vector of indices of all elements of array Y that are Truth (new 2026-10-18)

U+2A42

Enlist

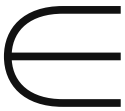


Enlist —; monadic ∈ Y: as defined in APL (deep flatten to vector)

U+220A

Sets and Membership: Dyadic

In



In — dyadic ∈ Y X yields Truth when X is in Y, Nil otherwise (J's e.). (new 2026-10-18)

U+2208

Bucket



Bucket — dyadic ∐ X Y: yields index of interval in X that Y falls into (new 2026-10-18)

U+2210

InLast



InLast — dyadic ∈̇ X Y: yields the index of the last occurrence of X in Y; yields length of Y if not found (new 2026-10-18)

U+22F5

InAt



InAt — dyadic ⋵ X Y: yields the index of Y whose value equals X; yields length of Y if not found (new 2026-10-18)

U+22F9

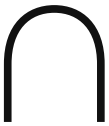
In



Enlist —; dyadic ∈ X Y: find/membership as defined in APL (new 2026-10-13)

U+220A

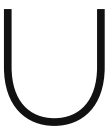
Intersection



Intersection —; dyadic ∩ A B: as in APL, elements in both (new 2026-10-14)

U+2229

Union



Union —; dyadic ∪ X Y: union as in APL (new 2026-10-14)

U+222A

Set Difference



Set Difference — dyadic ⍠A B: elements in A but not in B (new 2026-10-14)

U+2A43

Trains: Niladic

Reduce



Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Niladic / returns 0 (new 2026-10-12)

U+002F

Map



Map — ⟠ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Niladic ⟠ yields Truth, same as niladic 𝓘 (new 2026-10-12)

U+226C

Reduce (forward)



⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.

U+233F

Identity



Identity — 𝓘 V yields V unchanged; on a train, yields the train unchanged. Niladic 𝓘 yields Truth (new 2026-10-12)

U+1D574

Backstroke



Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Niladic \ yields 0 (new 2026-10-12)

U+005C

Stop



Stop — statement separator; separates statements on one line

U+22C4

Backstroke (forward)



Backstroke (forward) — Reverse Scan, paired with ⍀. ⍀ F V on a length-n V yields a length-n vector where element i is ⍀F (left fold) applied to the first i elements of V.

U+2340

Strip



Strip —; niladic: removes the train from head up to but not including the first Stop encountered

U+2ACF

Trains: Monadic

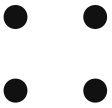
Meter

110



U+2214

Atomize



Atomize — monadic ∷ A: reads up to the first 32 elements of A (origin to end) and prepares them for use by train elements ahead (new 2026-10-14)

U+2237

Duplicate



Duplicate — copies a train to a train twice as long: ⊷ A B C → A B C A B C (was ==U+2A75, then ⁚ U+2236 — settled 2026-10-12)

U+22B7

Caboose



Caboose — picks up the train from head to Stop, then converts boxed contents to a vector (inverse of Atomize) (new 2026-10-15)

U+2A78



Meter

U+2214

Meter — measures its argument: ∔ V is ≺ + ⊷ V (zip the needle with the signal); yields M, a sequence of (needle, signal) pairs, needle ∈ -/0/+. M works in place of V in all Collate modes: the needle selects the function, the function transforms the signal (was ±, corrected to ≺ 2026-10-12)

Reduce



Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Monadic / V yields a linear array of the scalars ripped from V, highest to lowest indices.

U+002F

Map



Map — ⍁ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Monadic ⍁ B is identical to Identity 𝕴 B.

U+226C

Reduce (forward)



⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.; Monadic ⌿ V and niladic ⌿ behave identically to / (new 2026-10-12)

U+233F

Zero



Zero — on a train, replaces all values with zeroes while keeping shapes (new 2026-10-12)

U+1D585

Backstroke

113



U+005C

Duplicate Head



Duplicate Head — Ⅱ A ... yields A A ...; duplicates head of train, aka duplicate/dup (new 2026-10-14)

U+2161

Backstroke (forward)



Backstroke (forward) — Reverse Scan, paired with ⫻. ⟊ F V on a length-n V yields a length-n vector where element i is ⫻F (left fold) applied to the first i elements of V.; Monadic ⟊ V and niladic ⟊ behave identically to \ (new 2026-10-12)

U+2340

Pipeline



Pipeline —

U+2AF7



Backstroke

`U+005C`

Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Monadic \ V yields a linear array of the scalars ripped from V, lowest to highest indices.

Head



Head — sugar for ⊐1 (new 2026-10-15)

U+2991

Tail



Tail — sugar for ⊐-2 (new 2026-10-15)

U+2992

Duplicate Tail



Duplicate Tail — ⧟... A yields ... A A; duplicates tail of train (new 2026-10-13)

U+29DE

Strip

115



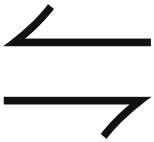
U+2ACF

⊐ Strip

U+2ACF

Strip —; monadic ⊐ X: removes the train from position X (0 is head) up to but not including the first Stop encountered; when X < 0, strips from position X in reverse order back to head, where −1 is the final tail position adjacent to the first Stop encountered (new 2026-10-15)

Swap



Swap — ⇋ A B ... yields B A ...; swaps head two of train. Single: ⇋ A Stop X yields X. Bare: ⇋ Stop X yields X (new 2026-10-13)

U+21CB

Replicate

117



U+232C

Rotate Head



Rotate Head — ⟲ A B C ... yields B C ... A; rotates head to tail (new 2026-10-13)

U+27F2

Rotate Tail



Rotate Tail — ⟳ ... A B C yields C ... A B; rotates tail to head (new 2026-10-13)

U+27F3



Replicate

U+232C

Replicate — ⌬ N X Stop yields X replicated N times; e.g. ⌬ 5 A Stop gives A A A A A. Zero case: ⌬ 0 A Stop X yields X. Negative N is antireplicate: ⌬ -1 A A A A A Stop yields A A A A. Empty antireplicate: ⌬ -1 Stop X yields X. Bare: ⌬ Stop X yields X (new 2026-10-13)

Nip



Nip — ↤ A B ... yields A ...; drops second, keeps head (new 2026-10-13)

U+21F7

Tuck



Tuck — ↦ A B ... yields A B A ...; copies head under second (new 2026-10-13)

U+21F8

Reverse Train



Reverse Train — reverses order of train (new 2026-10-13)

U+21F9

Over



Over — ⤺ A B ... yields B A B ...; copies second to head (new 2026-10-13)

U+293A

Triple

Ⅲ

Triple — Ⅲ X yields X X X; triplicates top of stack (new 2026-10-13)

U+2162

Fork



Fork — ⋔ F G H X: sugar for G H Swap F Dup X (J fork; new 2026-10-14)

U+22D4

Drop Train



Drop Train — sugar for reverse, replicate -1, reverse; drops head of train (new 2026-10-13)

U+291A

Trains: Dyadic

Applyself



Applyself — ⍣⍤1 f n a applies f n times to a; sugar for ⫷a ⍵⊂⍣⍺⍵ f n (from APL power; new 2026-10-12)

U+2363

Pipeline



Pipeline —; dyadic ⫷X Y applies vector Y, readable as a train of monadic functions, in sequence to X (new 2026-10-12)

U+2AF7

Functions: Monadic

Diction



Diction — dict-backed decision-table constructor (was Vectorize, renamed 2026-10-11). ¡ A B → {A:B}; ¡ A B C D → {A:B,C:D}; monadic ¡ (a b c) self-maps into a set. Membership sugar: X member Y → ⊤ (X ¿ (¡ Y))

U+00A1

Glyph



Glyph — the name for a single character value. Glyph ← ≡ ⌧: Glyph's function is the literalize train (via monadic ≡) — to make a Glyph, literalize

U+00AE

Decide

125



U+00BF

Function



Function — gets glyph del (∇, nabla): the function-definition operator. Monadic ∇ Y executes array Y as an FSA. (monadic 2026-10-19)

U+2207

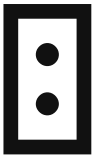


Decide

U+00BF

Decide — first-match generalized switch/match over n×2k train tables (was When, 2026-10-11). n×2 returns 1×1 (scalar); n×4 returns 1×2; generally n×2k returns 1×k. An explicit trailing C is unmatched-result filler, not an else branch; without filler, unmatched returns Nil without logging an error. Canonical: R ← X ¿ ¡ A B C

Print



Print — the quad-colon verb: prints the value from its right, then passes that value to its left (new 2026-10-11)

U+2360

(literalize)



Literalize next nonspace char, or ※0x???? by codepoint

U+2A33

Rail



Rail — monadic ⌐¯X yields a vector of symbols from train X (train-to-array conversion). (new 2026-10-19)

U+2ACD

Lay



Lay — monadic ¯¬V yields the train laid down from origin to limit of vector V (array-to-train, inverse of Rail). (new 2026-10-19)

U+2ACE

Lambda

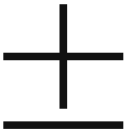


Lambda — lambda abstraction (new 2026-10-13)

U+039B

Functions: Dyadic

Collate



Collate — executes one of three trains upon a value: for less than zero, for zero, or for greater than zero (was AntiSignum ≺ U+22B0, renamed 2026-10-12; complement to signum)

U+00B1

Curry

130



U+1D49E

$\mathcal{C}$ Curry

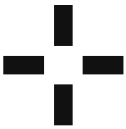
U+1D49E

Curry — intrinsic. Partial application: turns a dyadic verb into a monadic predicate by binding an argument, e.g. making = X from =. Semantics: Gets Q Curry P args: args to omega prime, omega prime shifts into alpha prime (binding the leftmost argument, textbook curry order), Q X makes X alpha, gets slice -1 of omega prime A, then call P; Q alpha yields P evaluated upon alpha prime, omega prime, and alpha as arguments in that order. Omega becomes nil upon Q's execution. Principle: Gets Q Curry P train is the principle for creating curries (Q is an example name assigned to the curried function, not a verb). (new 2026-10-18)

Errors: Niladic

Silence

134



U+22B9

Compress

135



U+22D6

Errline



Last error: which line

U+274C

Errmsg



Last error: what (type)

U+274E



Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Niladic ⊹ yields boxed Nil, a trial vector for enclosing operators to work on for starts (the dual of Compress; new 2026-10-12)



Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Niladic ⋖ yields boxed Truth, a trial vector for enclosing operators to work on for starts (a comprehension filter; new 2026-10-12)

Cut

Cut —; niladic: on error, erases the error and returns Nil

U+22EE

Debugger

✔⋄ *enters debugger iff error registers set*

U+2714

Errdmd

Last error: diamond position in line

U+2727

Succeed

Succeed — like Cut, except returns Truth (new 2026-10-15)

U+2999

Anticut



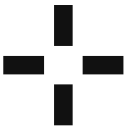
Anticut — on error throws to debugger; on return from debugger, returns Nil

U+22EF

Errors: Monadic

Silence

140



U+22B9

Compress

141



U+22D6

Cut



Cut —; monadic ⋮ X: checks X for Truth before returning

U+22EE



Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Monadic ⊹ V or ⊹ M means Z is identity: keep zeros (or zero-needle signals) as-is, drop nonzeros.



Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Monadic ⋖ V or ⋖ M means P is identity: pure filter, no transform.

Classes and Properties: Niladic

Constant

145

α

U+03B1

Epsilon

ε

Epsilon — holds the result of any getter, setter, or nabla when no explicit return glyph is used

U+03B5

Omega

ω

Omega — the vector of arguments passed to a setter (⊲)

U+03C9

α Constant

`U+03B1`

Constant — declares a class constant: X α name ← value names a constant for class X. α is also the original (pre-override) value: read-only, may not be changed directly; trying registers an error. As the original value α is an ad hoc constant — contextual to the setter, not a declared class constant

Classes and Properties: Monadic

Increment



Increment —; monadic ⊕ D: invokes D's setter with nabla + 1 as the xthrement.

U+2295

Setter



Setter — the setter nabla: defines a property setter, as ∇ defines a function. Generalized: takes an xthrement argument — modifies instance data (increment/decrement generalized) rather than assigning (new 2026-10-14)

U+22B2

Getter



Getter — the getter nabla: defines a property getter, as ∇ defines a function

U+22B3

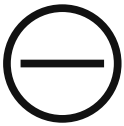
Property



Property —; Monadic ⋘ D: yields the result of D's getter (getter mode of D's class call) (new 2026-10-14)

U+22D8

Decrement



Decrement —; monadic ⊖ D: invokes D's setter with nabla - 1 as the xthrement.

U+2296

Divide By



Divide By —; monadic ⊘ D: invokes D's setter with nabla ÷ 2 as the xthrement (halve).

U+2298

Multiply By



Multiply By —; monadic ⊛ D: invokes D's setter with nabla × 2 as the xthrement (double).

U+229B

Classes and Properties: Dyadic

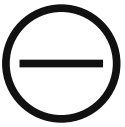
Increment



Increment —; Dyadic ⊕ D X: invokes D's setter with nabla + X as the xthrement (new 2026-10-14)

U+2295

Decrement



Decrement —; Dyadic ⊖ D X: invokes D's setter with nabla - X as the xthrement (new 2026-10-14)

U+2296

Multiply By



Multiply By —; Dyadic ⊛ D X: invokes D's setter with nabla × X as the xthrement (new 2026-10-14)

U+229B

Property



Property —; dyadic ⋘ I D: invokes D's getter (getter mode of D's class call), then invokes I's setter (setter mode of I's class call) with the result as argument; yields the assigned data as a pass-through.

U+22D8

Divide By



Divide By —; Dyadic ⊘ D X: invokes D's setter with nabla ÷ X as the xthrement (new 2026-10-14)

U+2298

(method binding)



Binds method to object: X⊡function

U+22A1

Gets



Gets — ⇔ A G: A stored in symbols[G], G stored in instances[A]; bidirectional binding (new 2026-10-13)

U+2B04

The Virtual: Niladic

IntegerBounded



IntegerBounded — bounded integer class: min/max enforced on creation, assignment, math; clamp + silent sticky flag on violation (new 2026-10-15)

U+012A

Trapdoor



Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).

U+1D21D

VirtualSpace



VirtualSpace — the VR coordinate space: navel origin, forward half-sphere, [r;θ;φ], angles ±32767, distance in mm (new 2026-10-15)

U+1D4E5

Prim



Prim — base VR primitive: name, position, collision machinery; posted via Virtual (new 2026-10-16)

U+1D7D8

Raster



Raster — VR prim ⊑ Prim: array of pels, rank 4 [Y,X,1,3] (new 2026-10-16)

U+1D7D9

Point



Point — VR prim ⊑ Prim: dot prim, [N,6] (H,V,D,RGB) (new 2026-10-16)

U+1D7DA

Triangle



Triangle — VR prim ⊑ Prim: 9 points × 2 thetas + D + RGB = 22 numbers, [N,22] (new 2026-10-16)

U+1D7DB

The Virtual: Monadic

Trapdoor

158



U+1D21D

⊂ Trapdoor

`U+1D21D`

Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).; Monadic ⊂ A: puts the address of A into the accumulator before calling. ⇔ X ⊂ ∇ assembly ∇: X gets the address of the compiled assembly script. If X is so bound, niladic X calls the code. ε maps to the accumulator. The call yields ε

Dungeon

Dungeon: Places

Where

@

Where the hero is — bound to '@', the roguelike player glyph. Where⊡Zee yields Where [-1 (last); Where⊡Wye yields Where [1 (midposition — middle of 3 coords); Where⊡Exx yields Where [0 (zero). INDICES START AT ZERO

U+0040

Dungeon

£

The Dungeon — literally the set of all levels

U+00A3

Screen

§

The Screen — intrinsic 20×60 array (16 play + 4 I/O rows)

U+00A7

Level

¶

Level — the pilcrow: a single dungeon level, member of £ Dungeon

U+00B6

Cell

¢

A dungeon cell — the cent sign joins the family: § £ ¢. Cell⊡location ← £ Where: the cell's location is the Dungeon at Where. A dungeon square IS a dictionary of instances (the ⓘ registry's entries present at that square)

U+00A2

Up



Up — the upward/north movement. Where⊡up yields ∅ when Where⊡Zee is 0, else Where with Zee decremented (refined 2026-10-10: boundary-checked — top edge is a quiet Nil, not a negative index)

U+25B2

Down



Down — the downward/south movement, counterpart to Up. Where⊡down yields ∅ when Where⊡Zee ≥ Dungeon⊡Zee (bottom edge), else Where⊡Zee + 1 (confirmed 2026-10-10)

U+25BC

Dungeon: Classes

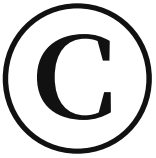
Class



Class — the circled dot. X⊡Class gets a RANK-2 ARRAY OF TRAINS: a class IS a decision table made of trains. Class instances are callable; the train defaults to Nil (∅)

U+2299

Classes



Classes — the set of all classes (plural of Class), parallel to £ Dungeon as the set of all levels. Implementation: a single twin (bidirectional word ↔ glyph) dictionary at global scope

U+24B8

Instances

166



U+24BE

Instance



Instance — the INDIVIDUAL instance (singular), distinct from Ⓘ the table. Joins the blackboard family (𝕏 𝕐 ℤ ℙ 𝕆). Carries the (glyph, hash-at-creation) identity tuple

U+1D540



Instances

U+24BE

Instances — the TABLE/set of all instances (plural), parallel to Ⓒ Classes. Implementation: a single twin (bidirectional word ↔ glyph) dictionary at global scope. Instance identity is the (glyph, hash-at-creation) tuple, used as the key field in both the global instance dictionary and each square (Cell) dictionary

Interpreter



Interpreter — the Wuxia interpreter (new 2026-10-15)

U+203D

Inherits



Inherits — the inheritance operator (square subset of)

U+2291

Isa



Isa — the is-a / instantiation operator (2026-10-10). Monadic ∈ Y: Y is a class to copy data from; new instance.ε = Y.ε. Dyadic ∈ Y X: X is a class, optional initializer; new instance.ε = X.ε (new 2026-10-14)

U+22FF

Object



Object — a DUNGEON object, not a generic OOP object. The blackboard O joins the projection family (𝕏 𝕐 ℤ ℙ): the root of the dungeon's object hierarchy, what every dungeon entity inherits from

U+1D546

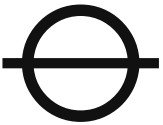
VirtualDistance



VirtualDistance — distance in VirtualSpace, 0-512 (new 2026-10-15)

U+27F0

Angle



Angle — a VR angle (theta/phi) (new 2026-10-15)

U+29B5

VirtualHorizontal



VirtualHorizontal — horizontal angle in VirtualSpace (new 2026-10-15)

U+29D3

VirtualVertical



VirtualVertical — vertical angle in VirtualSpace (new 2026-10-15)

U+29D7

Dungeon: Objects

Ring



Ring — the blackboard R: a dungeon object (ring), third of the dungeon entity glyphs

U+211D

Wand



Wand — left tack: a dungeon object (wand)

U+22A3

Armor



Armor — the blackboard A: a dungeon object (armor), first of the dungeon entity glyphs

U+1D538

Weapon



Weapon — the blackboard W: a dungeon object (weapon), second of the dungeon entity glyphs

U+1D54E

Amulet



Amulet — eight-petalled floret: a dungeon object (amulet). (Was U+1DF76, tofu on device — swapped 2026-10-10)

U+2741

Food



Food — the blackboard F: a dungeon object (food)

U+1D53D

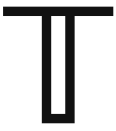
Scroll



Scroll — the blackboard S: a dungeon object (scroll)

U+1D54A

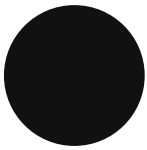
Tool



Tool — the blackboard T: a dungeon object (tool)

U+1D54B

Gem



Gem — solid black circle: a dungeon object (gem). (Was U+2733 — swapped 2026-10-10)

U+25CF

Potion



Potion — heavy exclamation: a dungeon object (potion)

U+2757

Dungeon: Entities

Player



Player — black smiling face: the dungeon's player entity. Inherits Monster plus a few extra fields Monster does not have

U+263B

Monster



Monster — the blackboard M: a dungeon monster, first of the monster glyphs

U+1D544

Dungeon: Food

Tin



Tin — circled white star: a dungeon food item (tin), first of the object species

U+272A

RefriedBeans



RefriedBeans — dark-encircled six: grants teleportation

U+277B

Spinach



Spinach — dark-encircled seven: Popeye strength

U+277C

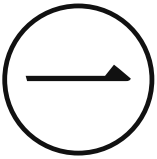
Sushi



Sushi — circled congratulation: the sushi/sashimi food category. (Was U+2722 — swapped 2026-10-11)

U+3297

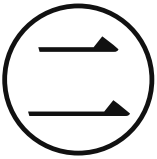
CarpSashimi



CarpSashimi — circled kanji: sushi item

U+3280

WhaleSashimi



WhaleSashimi — circled kanji: sushi item

U+3281

EelSashimi



EelSashimi — circled kanji: sushi item

U+3282

SalmonSashimi



SalmonSashimi — circled kanji: sushi item

U+3283

TunaSashimi



TunaSashimi — circled kanji: sushi item

U+3284

CrabSashimi



CrabSashimi — circled kanji: sushi item

U+3285

DragonSashimi



DragonSashimi — circled kanji: sushi item

U+3286

TurtleSashimi



TurtleSashimi — circled kanji: sushi item

U+3287

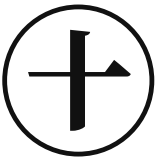
Oyster



Oyster — circled kanji: sushi item

U+3288

Uni



Uni — circled kanji: sushi item

U+3289

Gyuniku



Gyuniku — circled kanji: sushi item

U+328A

Asian



Asian — squared hand: the Asian food category. (Was U+328C — swapped 2026-10-11)

U+1F210

Ramen



Ramen — squared cjk unified ideograph-5b57: Asian food

U+1F211

FortuneCookie



FortuneCookie — squared cjk unified ideograph-53cc: Asian food

U+1F212

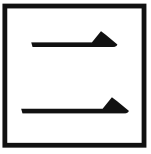
Kelp



Kelp — squared katakana de: Asian food

U+1F213

Dumplings



Dumplings — squared cjk unified ideograph-4e8c: Asian food

U+1F214

BowlOfRice



BowlOfRice — squared cjk unified ideograph-591a: Asian food

U+1F215

SpringRolls



SpringRolls — squared cjk unified ideograph-89e3: Asian food

U+1F216

Tofu



Tofu — squared cjk unified ideograph-5929: Asian food

U+1F217

PreservedEgg



PreservedEgg — squared cjk unified ideograph-4ea4: Asian food

U+1F218

Soup



Soup — circled twenty: the soup category

U+2473

Deli



Deli — black diamond minus white X: the delicatessen food category (the 25 staples, names lost 2026-10-10 — rebuild pending)

U+2756

Fruit



Fruit — domino tile back: the fruit food category

U+1F030

CantoneseDuck



CantoneseDuck — squared cjk unified ideograph-6620: Asian food

U+1F219

Egg

Ø

Egg — o with stroke: the egg. (Was U+2800 — swapped 2026-10-11)

U+00F8

Meat



Meat — CJK beautiful: the meat category

U+4E3D

Creature



Creature — o with long stroke overlay: the monster-shuffle category

U+A74B

Candy



Candy — mahjong joker: the candy category

U+1F02A

Goodbar



Goodbar — mahjong east wind: candy

U+1F000

Honee



Honee — mahjong south wind: candy

U+1F001

DRation



DRation — mahjong west wind: candy

U+1F002

DoveBar



DoveBar — mahjong north wind: candy

U+1F003

AcidBroth

❶

AcidBroth — dark-encircled number: soup

U+2776

GoldenSoup

❷

GoldenSoup — dark-encircled number: soup

U+2777

StonedSoup

❸

StonedSoup — dark-encircled number: soup

U+2778

CrowBar



CrowBar — mahjong winter: candy

U+1F029



MoldySoup

MoldySoup — dark-encircled number: soup

U+2779



Manwich

Manwich — dark-encircled number: soup

U+277A



CreamOfAlgaeSoup

CreamOfAlgaeSoup — dark-encircled number: soup

U+277E



ChickenNoodleSoup

ChickenNoodleSoup — dark-encircled number: soup

U+277F

Apple



Apple — domino tile: fruit, inherits Fruit

U+1F032

Orange



Orange — domino tile: fruit, inherits Fruit

U+1F033

Pear



Pear — domino tile: fruit, inherits Fruit

U+1F034

Melon



Melon — domino tile: fruit, inherits Fruit

U+1F035

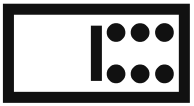
Banana



Banana — domino tile: fruit, inherits Fruit

U+1F036

Carrot



Carrot — domino tile: fruit, inherits Fruit

U+1F037

Fig



Fig — domino tile: fruit, inherits Fruit

U+1F038

Lemon



Lemon — domino tile: fruit, inherits Fruit

U+1F039

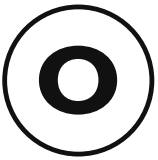
Tomb



Tomb — circled 10 on black square: safe-row tin, now Sundry

U+3248

Sundry



Sundry — circled ideograph: the sundry/miscellaneous category

U+3267

Lime



Lime — domino tile: fruit, inherits Fruit

U+1F03A

Grapes



Grapes — domino tile: fruit, inherits Fruit

U+1F03B

Pesto



Pesto — circled 20 on black square: safe-row tin, now Sundry

U+3249

KelpFrond



KelpFrond — circled water: sundry item (kelp is seaweed)

U+328C

EucalyptusLeaf



EucalyptusLeaf — circled wood: sundry item

U+328D

CandyBar



CandyBar — circled metal: sundry item

U+328E

LembasWafer



LembasWafer — circled earth: sundry item

U+328F

CramRation



CramRation — circled sun: sundry item

U+3290

FoodRation



FoodRation — circled medicine: sundry item

U+32A9

KRation



KRation — circled religion: sundry item

U+32AA

CloveOfGarlic



CloveOfGarlic — circled attention: sundry item

U+329F

SprigOfWolfsbane



SprigOfWolfsbane — circled item: sundry item

U+32A0

LumpOfRoyalJelly



LumpOfRoyalJelly — circled copy: sundry item

U+32A2

CRation



CRation — circled study: sundry item

U+32AB

Dungeon: Species inheritance

EarthElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+2295

FireElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+229B

AirElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+2A37

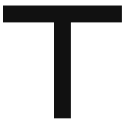
WaterElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+2A38

PurpleWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+22A4

BabyPurpleWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+22A6

LongWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+2A53

BabyLongWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+2A55

LargeKobold



⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+049C

Kobold



⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+049D

MasterMindFlayer



⊑ *Mindflayers — Species inherits its category (2026-10-10)*

U+04F4

MindFlayer



⊑ *Mindflayers — Species inherits its category (2026-10-10)*

U+04F5

StoneGiant

Ġ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+0120

Giant

Ģ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+0122

KoboldLord

Ќ

⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+040C

KoboldShaman

Ҝ

⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+049B

FireGiant

Ĝ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+011C

HillGiant

Ğ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+011E

FrostGiant

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⊑ *Giants — Species inherits its category (2026-10-10)*

U+0193

Ettin

G ˇ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+01E6

VampireBat

ƀ

⊑ *Vampires — Species inherits its category (2026-10-10)*

U+0180

StormGiant

Ǧ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+01E4

Vampire

Ỿ

⊑ *Vampires — Species inherits its category (2026-10-10)*

U+1E7F

Titan

⅁

⊑ *Giants — Species inherits its category (2026-10-10)*

U+2141

VampireLeader



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+1E7D

VampireLady



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

VampireLord



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

Nosferatu



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

HumanMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

ElfMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

DwarfMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

GnomeMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

OrcMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

KoboldMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

GiantMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

EttinMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

KoboldZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

GnomeZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

OrcZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

DwarfZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

ElfZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

HumanZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

GiantZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

EttinZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

Ghoul

g

⊑ *Undead — Species inherits its category (2026-10-10)*

U+210A

WoodElf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

GreenElf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

GreyElf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

BabyDragon

⁇

⊑ *Dragons — Species inherits its category (2026-10-10)*

U+2047

Elf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

ElfNoble



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2E19

ElvenMonarch

支

⊑ *Elves — Species inherits its category (2026-10-10)*

U+2F40

Centipede

¢

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00A2

CaveSpider

¤

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00A4

Scorpion

±

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00B1

Dragon

⁇

⊑ *Dragons — Species inherits its category (2026-10-10)*

U+2047

GiantSpider

Ø

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00D8

Mastodon

Ψ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+03A8

Mumak

Ψ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+03A8

Baluchitherium

ϒ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+03D2

Wumpus

¿

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+00BF

Trapper



⊑ *Beasts — Species inherits its category (2026-10-10)*

U+048A

Minotaur

Ṁ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+1E40

Titanothere



⊑ *Beasts — Species inherits its category (2026-10-10)*

U+213A

Bugbear

ö

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+00F6

Owlbear

ø

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+00F8

Jabberwock

ƹ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+01B8

Rothe

ȣ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+0222

Shade

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

Leocrotta

ꝉ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+023D

Stalker

Ǝ

⊑ *Lurkers — Species inherits its category (2026-10-10)*

U+042D

LurkerAbove



⊑ *Lurkers — Species inherits its category (2026-10-10)*

U+214F

Skeleton



⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

Foocubus



⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

MountainNymph



⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

WoodNymph



⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

WaterNymph

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

MerfolkRoyal

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

MerfolkNoble

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

MerfolkWitch

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

WizardOfYendor

¥

⊑ *Specials — Species inherits its category (2026-10-10)*

U+00A5

Shopkeeper

§

⊑ *Specials — Species inherits its category (2026-10-10)*

U+00A7

MerfolkCommon

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

GrandMaster

⁇

⊑ *Specials — Species inherits its category (2026-10-10)*

U+2047

Tin



⊑ *Food — Tin inherits Food — first object species joins the hierarchy (2026-10-10)*

U+272A

AcidBroth



⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2776

GoldenSoup



⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2777

StonedSoup



⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2778

MoldySoup

4

⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2779

Manwich

5

⊑ *Soup — Soup inherits Soup — the human-flesh tin (2026-10-10)*

U+277A

CreamOfAlgaeSoup

9

⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+277E

ChickenNoodleSoup

10

⊑ *Soup — Soup inherits Soup (2026-10-10)*

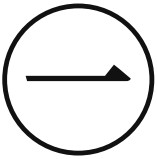
U+277F



RefriedBeans

⊑ *Soup — Soup item inherits Soup — grants teleportation (2026-10-11)*

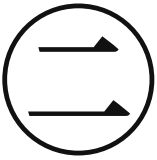
U+277B



CarpSashimi

⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3280



WhaleSashimi

⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3281



Sushi

⊑ *Food — Sushi inherits Food — the sushi food category (2026-10-10)*

U+3297

EelSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3282

SalmonSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3283

TunaSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3284

CrabSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3285

DragonSashimi



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3286

TurtleSashimi



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3287

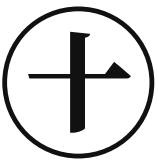
Oyster



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3288

Uni



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3289

Gyuniku



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+328A

Asian



⊑ *Food — Asian inherits Food — the Asian food category (2026-10-10)*

U+1F210

Ramen



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F211

FortuneCookie



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F212

Kelp



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F213

Dumplings



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F214

BowlOfRice



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F215

SpringRolls



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F216

Deli



⊑ *Food — Deli inherits Food — the delicatessen food category (2026-10-10)*

U+2756

Tofu



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F217

PreservedEgg



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F218

CantoneseDuck



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F219

KelpFrond



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328C

EucalyptusLeaf



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328D

CandyBar



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328E

LembasWafer



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328F

CramRation

㊐

⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+3290

FoodRation

㊕

⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A9

KRation

㊗

⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AA

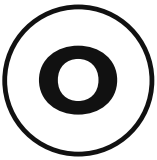
CRation

㊘

⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AB

Sundry



⊑ *Deli — Sundry is part of Deli (2026-10-10)*

U+3267

CloveOfGarlic



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+329F

SprigOfWolfsbane



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A0

LumpOfRoyalJelly



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A2

Fruit



⊑ *Food — Fruit inherits Food — the fruit food category (2026-10-10)*

U+1F030

Apple



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F032

Orange



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F033

Pear



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F034

Melon



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F035

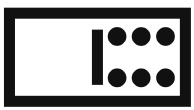
Banana



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F036

Carrot



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F037

Fig



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F038

Soup



⊑ *Tin — Soup inherits Tin — the soup category (2026-10-11)*

U+2473

Lemon



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F039

Lime



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03A

Grapes



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03B

Goodbar



⊑ *Candy — Goodbar inherits Candy (2026-10-11)*

U+1F000

Honee



⊑ *Candy — Honee inherits Candy (2026-10-11)*

U+1F001

DRation



⊑ *Candy — DRation inherits Candy (2026-10-11)*

U+1F002

Candy



⊑ *Food — Candy inherits Food — the candy category (2026-10-11)*

U+1F02A

Tomb



⊑ *Sundry — Tomb inherits Sundry*
(2026-10-11)

U+3248

Pesto



⊑ *Sundry — Pesto inherits Sundry*
(2026-10-11)

U+3249

DoveBar



⊑ *Candy — DoveBar inherits Candy*
(2026-10-11)

U+1F003

CrowBar



⊑ *Candy — CrowBar inherits Candy*
(2026-10-11)

U+1F029

Egg

Ø

⊑ *Creature — Egg inherits Creature (2026-10-11)*

U+00F8

Spinach



⊑ *Soup — Spinach (NetHack tin filling, Popeye strength) inherits Soup (2026-10-11)*

U+277C

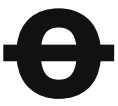
Meat



⊑ *Creature — Meat inherits Creature — the meat category (2026-10-11)*

U+4E3D

Creature



⊑ *Tin — Creature inherits Tin — the creature-meat tin category; the category that shuffles monsters as consumables (tins of monster meat) (2026-10-11)*

U+A74B



IntegerBounded

⊑ *Integer — IntegerBounded inherits Integer (new 2026-10-15)*

U+012A



IntegerBounded.max

⊑ *Integer — IntegerBounded.max is an Integer (new 2026-10-15)*

U+2047



IntegerBounded.min

⊑ *Integer — IntegerBounded.min is an Integer (new 2026-10-15)*

U+2047



IntegerBounded.epsilon

⊑ *Integer — IntegerBounded.epsilon is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.nabla

⁇

IntegerBounded.nabla enforces bounded gets and sets (new 2026-10-15)

U+2047

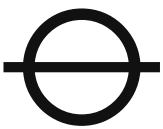
IntegerBounded.nabla

⁇

⊑ *Interpreter — IntegerBounded.nabla is an Interpreter (new 2026-10-15)*

U+2047

Angle



⊑ *IntegerBounded — Angle inherits IntegerBounded (new 2026-10-15)*

U+29B5

VirtualHorizontal



⊑ *Angle — VirtualHorizontal inherits Angle (new 2026-10-15)*

U+29D3

Egg

Ø

Egg stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)

U+00F8

VirtualDistance



⊑ *IntegerBounded — VirtualDistance inherits IntegerBounded (new 2026-10-15)*

U+27F0

VirtualVertical



⊑ *Angle — VirtualVertical inherits Angle (new 2026-10-15)*

U+29D7

Meat



Meat stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)

U+4E3D

Index

English Index

AcidBroth 188-1
AcidBroth 219-2
AirElemental 198-3
Amulet 172-1
Anagram 74-4
Anagram 81-4
Angle 168-2
Angle 237-3
Anticut 137-1
Antilog 28-2
Apple 190-1
Apple 230-2
Applyself 121-1
Approx 13-3
Arccosine 26-1
Arcsine 26-2
Arctangent 26-4
Armor 171-3
Asian 182-4
Asian 224-2
Atomize 109-2
BabyDragon 210-1
BabyLongWorm 199-4
BabyPurpleWorm 199-2
Backstroke 107-1
Backstroke 112-1
Backstroke (forward) 107-3
Backstroke (forward) 112-3
Baluchitherium 212-4
Banana 191-1
Banana 231-2
Base Ten Log 29-4
Bell 39-3
Bell 43-3
Bell 46-1
BowlOfRice 184-1
BowlOfRice 225-3
Box 86-1
Box 88-1
Box 89-1
Box 94-1
Bucket 102-2
Bugbear 214-1
Caboose 109-4
Candy 186-4
Candy 233-4
CandyBar 193-4
CandyBar 227-3
CantoneseDuck 185-4
CantoneseDuck 226-4
Cardinality 79-1
CarpSashimi 180-1
CarpSashimi 221-2
Carrot 191-2
Carrot 231-3
Catalogue 88-3
Catalogue 92-3
Catenate 72-2
Catenate 78-3
Catenate 82-1
CaveSpider 211-2
Ceiling 25-1
Ceiling 35-2
Cell 162-1
Center 79-2
Centipede 211-1
ChickenNoodleSoup 189-4
ChickenNoodleSoup 220-4
Class 165-1
Classes 165-2
CloveOfGarlic 195-1
CloveOfGarlic 229-2
Collate 129-1
Combinatoric 50-3
Combinatoric 53-1
Compress 133-2

Compress 139-2
Congruent 9-3
Congruent 11-3
Conjugate 21-3
Constant 144-1
Cosine 25-4
Count 75-3
CrabSashimi 181-2
CrabSashimi 222-4
CramRation 194-2
CramRation 228-1
CRation 195-4
CRation 228-4
CreamOfAlgaeSoup 189-3
CreamOfAlgaeSoup 220-3
Creature 186-3
Creature 235-4
CrowBar 188-4
CrowBar 234-4
Curry 129-2
Cut 136-1
Cut 139-3
Deal 44-1
Debugger 136-2
Decide 124-3
Decrement 148-1
Decrement 150-2
Deli 185-2
Deli 226-1
Determinant 66-1
Dice 40-4
Dice 43-4
Dice 46-2
Diction 124-1
Differentiate 56-3
Divide (Complex) 18-3
Divide (Complex) 23-3
Divide (Complex) 33-3
Divide (Polar) 18-4
Divide (Polar) 23-4
Divide (Polar) 33-4
Divide By 148-2
Divide By 151-1
Divisible 49-1
Divisible 52-2
Double 21-4
DoveBar 187-4
DoveBar 234-3
Down 162-3
Dragon 211-4
DragonSashimi 181-3
DragonSashimi 223-1
DRation 187-3
DRation 233-3
Drop 82-3
Drop Train 119-3
Dumplings 183-4
Dumplings 225-2
Dungeon 161-2
Duplicate 109-3
Duplicate Head 112-2
Duplicate Tail 114-3
DwarfMummy 205-3
DwarfZombie 207-4
EarthElemental 198-1
EelSashimi 180-3
EelSashimi 222-1
Egg 186-1
Egg 235-1
Egg 238-1
Elf 210-2
ElfMummy 205-2
ElfNoble 210-3
ElfZombie 208-1
ElvenMonarch 210-4
Enlist 100-1
Enumerate 75-1
Epsilon 144-2
Equality 12-1
Errdmd 136-3
Errline 133-3
Errmsg 133-4
Ettin 202-4
EttinMummy 206-4

EttinZombie 208-4
EucalyptusLeaf 193-3
EucalyptusLeaf 227-2
Euler 40-3
Evaluate 59-4
Extend 72-1
Extend 78-2
Extend 83-2
Exx 71-3
Factor 50-2
Factorial 19-1
Factorial 29-1
Factorial 35-1
Fetch 92-2
Fig 191-3
Fig 231-4
FireElemental 198-2
FireGiant 202-1
FixedPoint 60-1
Floor 25-2
Floor 35-3
Foocubus 216-2
Food 172-2
FoodRation 194-3
FoodRation 228-2
forall 13-2
Fork 119-2
Fornone 14-2
Forsome 14-1
FortuneCookie 183-2
FortuneCookie 224-4
Fourier 29-2
FrostGiant 202-3
Fruit 185-3
Fruit 230-1
Function 124-4
Gamma 40-1
GCD 52-3
Gem 173-1
Gets 151-3
Getter 147-3
Ghoul 209-1
Giant 201-2
GiantMummy 206-3
GiantSpider 212-1
GiantZombie 208-3
Glyph 124-2
GnomeMummy 205-4
GnomeZombie 207-2
Golden 40-2
GoldenSoup 188-2
GoldenSoup 219-3
Goodbar 187-1
Goodbar 233-1
Grade Down 72-4
Grade Up 72-3
GrandMaster 218-4
Grapes 192-4
Grapes 232-4
Greater Or Equal 12-4
Greater Than 11-2
GreenElf 209-3
GreyElf 209-4
Gyuniku 182-3
Gyuniku 224-1
Half 39-1
Half 43-1
Head 114-1
HillGiant 202-2
Honee 187-2
Honee 233-2
HumanMummy 205-1
HumanZombie 208-2
Hyperbolic Arccosine 28-4
Hyperbolic Arcsine 27-3
Hyperbolic Arctangent 28-3
Hyperbolic Cosine 27-2
Hyperbolic Sine 27-1
Hyperbolic Tangent 27-4
Identity 106-4
Implies 13-1
In 102-1
In 103-1
InAt 102-4

Increment 147-1
Increment 150-1
Indivisible 49-2
Indivisible 52-4
Inherits 167-2
InLast 102-3
Inner Product 63-2
Inner Product 65-2
Inner Product 68-2
InOf 99-3
Instance 165-4
Instances 165-3
Integer Division 59-3
IntegerBounded 154-1
IntegerBounded 236-1
IntegerBounded.epsilon 236-4
IntegerBounded.max 236-2
IntegerBounded.min 236-3
IntegerBounded.nabla 237-1
IntegerBounded.nabla 237-2
Integrate 56-4
Interpreter 167-1
Intersection 97-1
Intersection 99-1
Intersection 103-2
InverseFourier 29-3
Invert 9-1
Isa 167-3
IsAt 11-4
Jabberwock 214-3
Kelp 183-3
Kelp 225-1
KelpFrond 193-2
KelpFrond 227-1
Kobold 200-2
KoboldLord 201-3
KoboldMummy 206-2
KoboldShaman 201-4
KoboldZombie 207-1
Kount 75-4
KRation 194-4
KRation 228-3
L Factor 65-4
Lambda 127-1
Laminate 74-3
Laminate 81-3
LargeKobold 200-1
Lay 126-4
LCM 52-1
LembasWafer 194-1
LembasWafer 227-4
Lemon 191-4
Lemon 232-2
Leocrotta 215-2
Less Or Equal 12-3
Less Than 11-1
Level 161-4
Lime 192-3
Lime 232-3
Link 92-4
(literalize) 126-2
Local 57-3
LongWorm 199-3
LumpOfRoyalJelly 195-3
LumpOfRoyalJelly 229-4
LurkerAbove 215-4
Magnitude 30-1
Manwich 189-2
Manwich 220-2
Map 106-2
Map 111-2
MasterMindFlayer 200-3
Mastodon 212-2
Meat 186-2
Meat 235-3
Meat 238-4
Melon 190-4
Melon 231-1
MerfolkCommon 218-3
MerfolkNoble 217-3
MerfolkRoyal 217-2
MerfolkWitch 217-4
Meter 109-1
(method binding) 151-2

MindFlayer 200-4
Minotaur 213-3
Minus (Complex) 18-1
Minus (Complex) 23-1
Minus (Complex) 33-1
MoldySoup 189-1
MoldySoup 220-1
Monster 176-2
MountainNymph 216-3
Multiply 19-4
Multiply 24-4
Multiply 34-4
Multiply By 148-3
Multiply By 150-3
Multiply Matrices 63-3
Multiply Matrices 65-3
Multiply Matrices 68-3
Mumak 212-3
Natural Log 28-1
Natural Log 35-4
Negate 21-2
Negate / Subtract 17-2
NextLocal 56-1
Nil 7-2
Nip 118-1
Nosferatu 204-4
Not 9-2
Object 167-4
Omega 144-3
Optimize 57-2
Optimize 59-2
Orange 190-2
Orange 230-3
OrcMummy 206-1
OrcZombie 207-3
Outer Product 63-1
Outer Product 65-1
Outer Product 68-1
Over 118-4
Owlbear 214-2
Oyster 182-1
Oyster 223-3
Pear 190-3
Pear 230-4
Pee 71-1
Percent 19-2
Percent 24-1
Percent 34-1
Permute 74-2
Permute 81-2
Pesto 193-1
Pesto 234-2
Pi 39-2
Pick 41-1
Pick 43-2
Pipeline 112-4
Pipeline 121-2
Player 176-1
Plus 32-1
Plus (Complex) 17-3
Plus (Complex) 22-3
Plus (Complex) 32-3
Point 155-2
Potion 173-2
PreservedEgg 184-4
PreservedEgg 226-3
Prim 154-4
Prime 49-4
Prime 50-1
Primes 49-3
Print 126-1
Property 147-4
Property 150-4
PurpleWorm 199-1
Put 81-1
Quotient 24-3
Quotient 34-3
Rail 126-3
Ramen 183-1
Ramen 224-3
Rank 74-1
Raster 155-1
Ravel 77-2
RavelRows 77-1

Raze 88-4
Real 39-4
Reduce 106-1
Reduce 111-1
Reduce (forward) 106-3
Reduce (forward) 111-3
Reflect 75-2
RefriedBeans 179-2
RefriedBeans 221-1
Replicate 116-2
Reveal 59-1
Reverse 77-3
Reverse 82-4
Reverse Train 118-3
Ring 171-1
Roots 56-2
Rotate 78-4
Rotate 83-3
Rotate Head 116-3
Rotate Tail 116-4
Rothe 214-4
SalmonSashimi 180-4
SalmonSashimi 222-2
Scorpion 211-3
Screen 161-3
Scroll 172-3
Set Difference 103-4
Setter 147-2
Shade 215-1
Shape 78-1
Shape 83-1
Shopkeeper 218-2
Sight 79-3
Signum 21-1
Signum / Plus 17-1
Silence 133-1
Silence 139-1
Sine 25-3
Skeleton 216-1
Solve 57-1
Soup 185-1
Soup 232-1
Spinach 179-3
Spinach 235-2
SprigOfWolfsbane 195-2
SprigOfWolfsbane 229-3
SpringRolls 184-2
SpringRolls 225-4
Square 19-3
Square 36-1
Square Root 22-2
Stalker 215-3
Star 24-2
Star 34-2
StonedSoup 188-3
StonedSoup 219-4
StoneGiant 201-1
Stop 107-2
StormGiant 203-2
Strip 107-4
Strip 114-4
Subtract 32-2
Succeed 136-4
Sum 22-1
Summarize 86-3
Summarize 89-3
Summarize 92-1
Sundry 192-2
Sundry 229-1
Sushi 179-4
Sushi 221-4
Swap 116-1
Tail 114-2
Take 82-2
Tangent 26-3
Times (Complex) 17-4
Times (Complex) 22-4
Times (Complex) 32-4
Times (Polar) 18-2
Times (Polar) 23-2
Times (Polar) 33-2
Tin 179-1
Tin 219-1
Titan 203-4

Titanothere 213-4
Tofu 184-3
Tofu 226-2
Tomb 192-1
Tomb 234-1
Tool 172-4
Transpose 77-4
Trapdoor 154-2
Trapdoor 157-1
Trapper 213-2
Triangle 155-3
Triple 119-1
Truth 7-3
Tuck 118-2
TunaSashimi 181-1
TunaSashimi 222-3
TurtleSashimi 181-4
TurtleSashimi 223-2
U Factor 66-2
UnApprox 13-4
Unbox 86-2
Unbox 88-2
Unbox 89-2
Unbox 94-2
Unequal 12-2
Uni 182-2
Uni 223-4
Union 97-2
Union 99-2
Union 103-3
Up 162-2
Vampire 203-3
VampireBat 203-1
VampireLady 204-2
VampireLeader 204-1
VampireLord 204-3
VirtualDistance 168-1
VirtualDistance 238-2
VirtualHorizontal 168-3
VirtualHorizontal 237-4
VirtualSpace 154-3
VirtualVertical 168-4
VirtualVertical 238-3
Wand 171-2
WaterElemental 198-4
WaterNymph 217-1
Weapon 171-4
WhaleSashimi 180-2
WhaleSashimi 221-3
Where 99-4
Where 161-1
WizardOfYendor 218-1
WoodElf 209-2
WoodNymph 216-4
Wumpus 213-1
Wuxia 7-1
Wye 71-4
Zee 71-2
Zero 111-4
Zip 89-4

Glyph Index

! (U+0021) 19-1, 29-1, 35-1
% (U+0025) 19-2, 24-1, 34-1
& (U+0026) 57-1
* (U+002A) 24-2, 34-2
+ (U+002B) 17-1, 21-1, 32-1
- (U+002D) 17-2, 21-2, 32-2
/ (U+002F) 106-1, 111-1
< (U+003C) 11-1
= (U+003D) 12-1
> (U+003E) 11-2
? (U+003F) 44-1
@ (U+0040) 161-1, 216-1, 216-2, 216-3, 216-4, 217-1
\ (U+005C) 107-1, 112-1
~ (U+007E) 9-1
¡ (U+00A1) 124-1
¢ (U+00A2) 162-1, 211-1
£ (U+00A3) 161-2
¤ (U+00A4) 211-2
¥ (U+00A5) 218-1
§ (U+00A7) 161-3, 218-2
¬ (U+00AC) 9-2
® (U+00AE) 124-2
± (U+00B1) 129-1, 211-3
¶ (U+00B6) 161-4
½ (U+00BD) 39-1, 43-1
¿ (U+00BF) 124-3, 213-1
Ø (U+00D8) 212-1
ö (U+00F6) 214-1
÷ (U+00F7) 24-3, 34-3
ø (U+00F8) 161-1, 214-2, 235-1, 238-1
Ĝ (U+011C) 202-1
ğ (U+011E) 202-2
Ġ (U+0120) 201-1
Ģ (U+0122) 201-2
Ī (U+012A) 154-1, 236-1
ƀ (U+0180) 203-1
Ɠ (U+0193) 202-3
Ƹ (U+01B8) 214-3
Ǧ (U+01E4) 203-2
Ǧ ˇ (U+01E6) 202-4
Ȣ (U+0222) 214-4
Ƚ (U+023D) 215-2
Λ (U+039B) 127-1
Ψ (U+03A8) 212-2, 212-3
α (U+03B1) 144-1
γ (U+03B3) 40-1
ε (U+03B5) 144-2
π (U+03C0) 39-2
ρ (U+03C1) 78-1, 83-1
φ (U+03C6) 40-2
ω (U+03C9) 144-3
ϒ (U+03D2) 212-4
Ϝ (U+03DC) 29-2
Ќ (U+040C) 201-3
Э (U+042D) 215-3
Ѫ (U+046A) 205-1, 205-2, 205-3, 205-4, 206-1, 206-2, 206-3, 206-4
Й̦ (U+048A) 213-2
ҍ (U+049B) 201-4
Ҝ (U+049C) 200-1
ҝ (U+049D) 200-2
Ӌ (U+04F4) 200-3
ӌ (U+04F5) 200-4
Ṁ (U+1E40) 213-3
ṽ (U+1E7D) 204-1
ỿ (U+1E7F) 203-3
⁽ (U+203D) 167-1
⁇ (U+2047) 204-3, 204-4, 210-1, 211-4, 217-2, 217-3, 217-4, 218-3, 218-4, 236-2, 236-3, 236-4, 237-1, 237-2
Ɛ (U+2107) 40-3
𝑔 (U+210A) 209-1
ℓ (U+2113) 56-1
№ (U+2116) 75-1
ℙ (U+2119) 71-1

ℜ	(U+211C)	59-1
ℝ	(U+211D)	171-1
℞	(U+211E)	56-2
ℤ	(U+2124)	70-1, 70-2, 207-1, 207-2, 207-3, 207-4, 208-1, 208-2, 208-3, 208-4
℩	(U+2129)	29-3
℧	(U+2127)	213-4
ℾ	(U+213E)	72-1, 78-2, 83-2
ℿ	(U+213F)	72-2, 78-3, 82-1
⅁	(U+2141)	203-4
⅋	(U+214B)	57-2, 59-2
⅏	(U+214F)	215-4
Ⅱ	(U+2161)	112-2
Ⅲ	(U+2162)	119-1
↑	(U+2191)	82-2
↓	(U+2193)	82-3
⇋	(U+21CB)	116-1
⇒	(U+21D2)	13-1
↤	(U+21F7)	118-1
↠	(U+21F8)	118-2
↮	(U+21F9)	118-3
∀	(U+2200)	13-2
∂	(U+2202)	56-3
∃	(U+2203)	14-1
∄	(U+2204)	14-2
∅	(U+2205)	7-2
∇	(U+2207)	124-4
∈	(U+2208)	102-1
ϵ	(U+220A)	100-1, 103-1
∐	(U+2210)	79-1, 102-2
∑	(U+2211)	22-1
∔	(U+2214)	109-1
√	(U+221A)	22-2
∩	(U+2229)	97-1, 99-1, 103-2
∪	(U+222A)	97-2, 99-2, 103-3
∫	(U+222B)	56-4
∷	(U+2237)	109-2
÷	(U+223B)	75-2
≈	(U+2248)	13-3
≉	(U+2249)	13-4
≖	(U+2256)	88-1
≗	(U+2257)	88-2
≠	(U+2260)	12-2
≡	(U+2261)	9-3, 11-3
≣	(U+2263)	11-4
≤	(U+2264)	12-3
≥	(U+2265)	12-4
≬	(U+226C)	106-2, 111-2
⊂	(U+2282)	86-1, 89-1, 94-1
⊃	(U+2283)	86-2, 89-2, 94-2
⊑	(U+2291)	167-2
⊕	(U+2295)	147-1, 150-1, 198-1
⊖	(U+2296)	148-1, 150-2
⊗	(U+2297)	63-1, 65-1, 68-1
⊘	(U+2298)	148-2, 151-1
⊙	(U+2299)	165-1
⊚	(U+229A)	86-3, 89-3, 92-1
⊛	(U+229B)	148-3, 150-3, 198-2
⊞	(U+229E)	79-2
⊠	(U+22A0)	19-3, 36-1
⊡	(U+22A1)	151-2
⊣	(U+22A3)	171-2
⊤	(U+22A4)	7-3, 199-1
⊢	(U+22A6)	199-2
⊶	(U+22B0)	89-4
⊲	(U+22B2)	147-2
⊳	(U+22B3)	147-3
⊷	(U+22B6)	21-3
⊸	(U+22B7)	109-3
⊹	(U+22B8)	92-2
⊹	(U+22B9)	133-1, 139-1
⋄	(U+22C4)	107-2
⋅	(U+22C5)	63-2, 65-2, 68-2
⋈	(U+22C8)	63-3, 65-3, 68-3
⋔	(U+22D4)	119-2
⋕	(U+22D5)	74-1
⋖	(U+22D6)	133-2, 139-2
⋘	(U+22D8)	147-4, 150-4
⋮	(U+22EE)	136-1, 139-3
⋯	(U+22EF)	137-1
⋲	(U+22F2)	77-1
⋳	(U+22F3)	77-2
⋵	(U+22F5)	102-3
⋸	(U+22F8)	99-3

⋹	(U+22F9)	102-4
⊏	(U+22FF)	167-3
⌈	(U+2308)	25-1, 35-2
⌊	(U+230A)	25-2, 35-3
⌬	(U+232C)	116-2
⌼	(U+233C)	66-1
⌽	(U+233D)	77-3, 82-4
⌿	(U+233F)	106-3, 111-3
⍀	(U+2340)	107-3, 112-3
⍉	(U+2349)	77-4
⍋	(U+234B)	72-3
⍒	(U+2352)	72-4
⍗	(U+2357)	30-1
⍵	(U+235B)	18-1, 23-1, 33-1
⍰	(U+2360)	126-1
⍣	(U+2363)	121-1
⍳	(U+2373)	75-3
⍷	(U+237C)	28-1, 35-4
⎈	(U+2388)	79-3
⎊	(U+238A)	52-1
⎢	(U+23A2)	49-1, 52-2
⏀	(U+23C0)	50-1
⑳	(U+2473)	185-1, 232-1
Ⓒ	(U+24B8)	165-2
Ⓘ	(U+24BE)	165-3
▲	(U+25B2)	162-2
▼	(U+25BC)	162-3
●	(U+25CF)	173-1
◝	(U+25DD)	25-3
◞	(U+25DE)	25-4
◶	(U+25F6)	26-1
◷	(U+25F7)	26-2
☘	(U+2618)	208-2, 209-3, 209-4, 210-2
☻	(U+263B)	176-1
✔	(U+2714)	136-2
✧	(U+2727)	136-3
✪	(U+272A)	179-1, 219-1
❁	(U+2741)	172-1
❌	(U+274C)	133-3
❎	(U+274E)	133-4
❖	(U+2756)	185-2, 226-1
❗	(U+2757)	173-2
❶	(U+2776)	188-1, 219-2
❷	(U+2777)	188-2, 219-3
❸	(U+2778)	188-3, 219-4
❹	(U+2779)	189-1, 220-1
❺	(U+277A)	189-2, 220-2
❻	(U+277B)	179-2, 221-1
❼	(U+277C)	179-3, 235-2
❾	(U+277E)	189-3, 220-3
❿	(U+277F)	189-4, 220-4
⟒	(U+27D2)	50-2
⟪	(U+27DA)	81-1
⟰	(U+27F0)	168-1, 238-2
⟲	(U+27F2)	116-3
⟳	(U+27F3)	116-4
⤚	(U+291A)	119-3
⤺	(U+293A)	118-4
⦃	(U+2980)	52-3
⦑	(U+2991)	114-1
⦒	(U+2992)	114-2
⦙	(U+2999)	136-4
⦵	(U+29B5)	168-2, 237-3
⦶	(U+29BD)	78-4, 83-3
⧊	(U+29CA)	28-2
⧓	(U+29D3)	168-3, 237-4
⧗	(U+29D7)	168-4, 238-3
⧾	(U+29DE)	114-3
⨂	(U+2A02)	18-2, 23-2, 33-2
⨃	(U+2A03)	41-1, 43-2
⨊	(U+2A0A)	74-2, 81-2
⨢	(U+2A22)	17-3, 22-3, 32-3
⨣	(U+2A23)	75-4
⨧	(U+2A27)	21-4
⨬	(U+2A2C)	18-3, 23-3, 33-3
⨯	(U+2A2F)	19-4, 24-4, 34-4
⨰	(U+2A30)	17-4, 22-4, 32-4
⨳	(U+2A33)	126-2
⨷	(U+2A37)	88-3, 92-3, 198-3
⨸	(U+2A38)	18-4, 23-4, 33-4, 198-4
⩅	(U+2A42)	99-4
⩏	(U+2A43)	103-4
⩍	(U+2A4D)	28-3
⩻	(U+2A53)	199-3

⩕	(U+2A55)	199-4
⩞	(U+2A5E)	74-3, 81-3
⩟	(U+2A5F)	29-4
⩩	(U+2A69)	50-3, 53-1
⩸	(U+2A78)	109-4
⪦	(U+2AA6)	27-1
⪧	(U+2AA7)	27-2
⪭	(U+2ABD)	27-3
⪮	(U+2ABE)	28-4
⫍	(U+2ACD)	126-3
⫎	(U+2ACE)	126-4
⫏	(U+2ACF)	107-4, 114-4
⫑	(U+2AD1)	27-4
⫓⊂	(U+2AD7)	26-3
⫓⪅	(U+2AD8)	26-4
⫩	(U+2AE9)	92-4
⫮	(U+2AEE)	49-2, 52-4
⫵	(U+2AF5)	88-4
⫷	(U+2AF7)	112-4, 121-2
⫽	(U+2AFD)	59-3
⬄	(U+2B04)	151-3
〻	(U+2E19)	210-3
支	(U+2F40)	210-4
㉘	(U+3248)	192-1, 234-1
㉙	(U+3249)	193-1, 234-2
㉧	(U+3267)	192-2, 229-1
㊀	(U+3280)	180-1, 221-2
㊁	(U+3281)	180-2, 221-3
㊂	(U+3282)	180-3, 222-1
㊃	(U+3283)	180-4, 222-2
㊄	(U+3284)	181-1, 222-3
㊅	(U+3285)	181-2, 222-4
㊆	(U+3286)	181-3, 223-1
㊇	(U+3287)	181-4, 223-2
㊈	(U+3288)	182-1, 223-3
㊉	(U+3289)	182-2, 223-4
㊊	(U+328A)	182-3, 224-1
㊌	(U+328C)	193-2, 227-1
㊍	(U+328D)	193-3, 227-2
㊎	(U+328E)	193-4, 227-3
㊏	(U+328F)	194-1, 227-4
㊐	(U+3290)	194-2, 228-1
㊗	(U+3297)	179-4, 221-4
㊞	(U+329F)	195-1, 229-2
㊠	(U+32A0)	195-2, 229-3
㊤	(U+32A2)	195-3, 229-4
㊩	(U+32A9)	194-3, 228-2
㊪	(U+32AA)	194-4, 228-3
㊫	(U+32AB)	195-4, 228-4
丽	(U+4E3D)	186-2, 235-3, 238-4
侠	(U+4FA0)	7-1
ⱺ	(U+A74B)	186-3, 235-4
⍵	(U+AB44)	60-1
𝐂	(U+1D21D)	154-2, 157-1
𝒜	(U+1D49C)	74-4, 81-4
𝒞	(U+1D49E)	129-2
𝒫	(U+1D4AB)	49-3
𝒱	(U+1D4E5)	154-3
𝔏	(U+1D50F)	57-3
𝔓	(U+1D513)	49-4
𝔸	(U+1D538)	171-3
𝔽	(U+1D53D)	172-2
𝕀	(U+1D540)	165-4
𝕄	(U+1D544)	176-2
𝕆	(U+1D546)	167-4
𝕊	(U+1D54A)	172-3
𝕋	(U+1D54B)	172-4
𝕎	(U+1D54E)	171-4
𝕏	(U+1D54F)	71-3
𝕐	(U+1D550)	71-4
𝔅	(U+1D56D)	39-3, 43-3, 46-1
𝔇	(U+1D56F)	40-4, 43-4, 46-2
𝔍	(U+1D574)	106-4
𝔏	(U+1D577)	65-4
𝔑	(U+1D57D)	39-4
𝔘	(U+1D580)	66-2
𝔷	(U+1D585)	111-4
𝖋	(U+1D58B)	59-4
𝟘	(U+1D7D8)	154-4
𝟙	(U+1D7D9)	155-1
𝟚	(U+1D7DA)	155-2
𝟛	(U+1D7DB)	155-3
🀀	(U+1F000)	187-1, 233-1
🀁	(U+1F001)	187-2, 233-2

	(U+1F002)	187-3, 233-3		(U+1F03A)	192-3, 232-3
	(U+1F003)	187-4, 234-3		(U+1F03B)	192-4, 232-4
	(U+1F029)	188-4, 234-4	手	(U+1F210)	182-4, 224-2
	(U+1F02A)	186-4, 233-4	字	(U+1F211)	183-1, 224-3
	(U+1F030)	185-3, 230-1	双	(U+1F212)	183-2, 224-4
	(U+1F032)	190-1, 230-2	デ	(U+1F213)	183-3, 225-1
	(U+1F033)	190-2, 230-3	二	(U+1F214)	183-4, 225-2
	(U+1F034)	190-3, 230-4	多	(U+1F215)	184-1, 225-3
	(U+1F035)	190-4, 231-1	解	(U+1F216)	184-2, 225-4
	(U+1F036)	191-1, 231-2	天	(U+1F217)	184-3, 226-2
	(U+1F037)	191-2, 231-3	交	(U+1F218)	184-4, 226-3
	(U+1F038)	191-3, 231-4	映	(U+1F219)	185-4, 226-4
	(U+1F039)	191-4, 232-2			